IRREGULAR GALAXY

Small Magellanic Cloud

CATALOG NUMBER NGC 292

DISTANCE 200,000 light-years

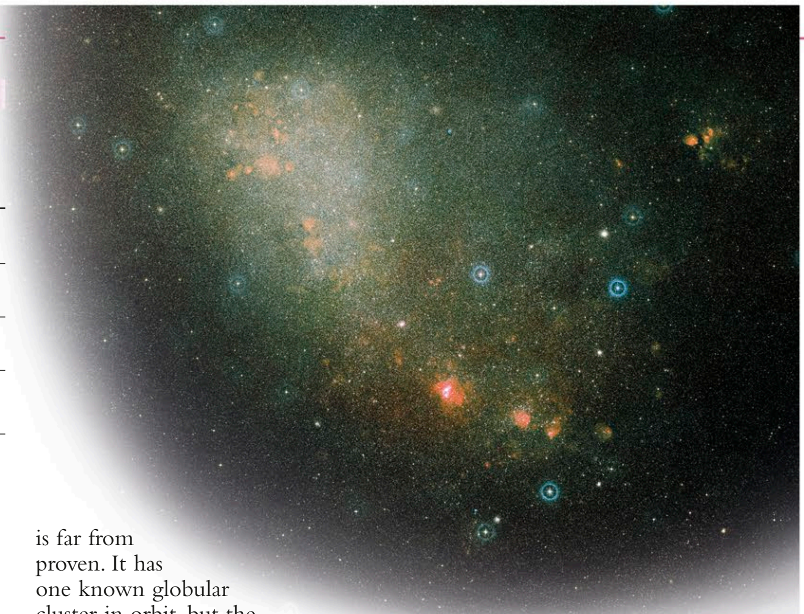
DIAMETER 10,000 light-years

MAGNITUDE 2.3

TUCANA

Like the Large Magellanic Cloud, the Small Magellanic Cloud (SMC) is an irregular galaxy in orbit around the Milky Way. It was in the SMC that Henrietta Leavitt discovered the Cepheid variable stars that were to unlock the secrets of the galactic distance scale (see pp.282, 356). Thanks to her discovery, astronomers know that the SMC is both more distant and genuinely smaller than the LMC, with around one-tenth of the larger cloud's mass. Like the LMC, the small cloud is also undergoing intense star formation. Some astronomers argue that the SMC also shows signs of a central barlike structure, but the case is far from proven. It has one known globular cluster in orbit, but the SMC lies deceptively close in the sky to one of the Milky Way's largest globulars—47 Tucanae.

Both of the Magellanic Clouds are ultimately doomed to be torn to shreds and absorbed into our own galaxy. They have survived several close passes of the Milky Way, but now share their orbit with a trail of gas, dust, and stars torn away during previous encounters. This "Magellanic Stream" has allowed astronomers to trace and refine their models for the orbits of the clouds.



CLOUD OF STARS
The SMC forms a distinctive wedge-shaped cloud in southern skies. The pinkish areas in this optical photograph show the galaxy's major star-forming regions.

EXPLORING SPACE

MAGELLAN'S DISCOVERY

The southernmost sky was not visible to Europeans until they visited the Southern Hemisphere. The Portuguese explorer Ferdinand Magellan was among the first to do so during his around-the-world voyage of 1519–1521. He was the first European to record two isolated patches of the Milky Way, which were later named after him.

FERDINAND MAGELLAN

CLOUD DETAILS
A near-infrared image of the LMC taken by the ESO's VISTA telescope reveals previously hidden stars tracing faint patterns that may indicate multiple spiral arms.

SC SPIRAL GALAXY

Triangulum Galaxy

CATALOG NUMBERS M33, NGC 598

DISTANCE 3 million light-years

DIAMETER 50,000 light-years

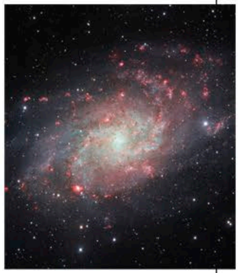
MAGNITUDE 5.7

TRIANGULUM

After the Andromeda Galaxy and the Milky Way, the Triangulum Galaxy (M33) is the third major member of the Local Group of galaxies. It is slightly more distant than the larger and brighter Andromeda Galaxy (M31), and the two lie close to each other in the sky. M33 is affected by its larger neighbor's gravity, and it may even be in a long, slow orbit around the giant Andromeda spiral.

Seen from Earth, M33 is fainter and more diffuse than M31—partly because it is closer to face-on than edge-on, and partly because it really is less spectacular. However, the Triangulum Galaxy is more typical of spiral galaxies than its unusually bright companions. As with several Local Group galaxies, M33 is large and bright enough in the sky for its features to be cataloged, and several of them have NGC numbers. Most prominent is the star-forming region NGC 604, the largest emission nebula known. At 1,500 light-years across, it dwarfs anything in our own galaxy.

FLOCCULENT SPIRAL
M33 is an example of a flocculent spiral—a galaxy with arms that divide like split ends and separate into patches. The clumpy star clouds are thought to form due to localized changes in density.



NEBULA NGC 604
The most massive stars of NGC 604 are so hot that their fierce stellar winds are helping shape the nebula in the Triangulum Galaxy. A visible-light image (purple) captures NGC 604's filaments of gas and dust, while X-ray data (blue) reveals expanding bubbles of superhot gas between them.